The Geometry Behind the Bessel Lattice: Twisted de Rham Cohomology, the Quantum Cohomology of $\mathbb{P}^1$, and the Meixner–Pollaczek Matching Matrices

Robin Langer and Claude Code

August 2026

Abstract

The forward matching matrix of the Meixner–Pollaczek dual pair (`meixner-pollaczek2.tex`) is integral over the rank-two lattice $\mathbb{Z}A \oplus \mathbb{Z}B$ with $A = \sqrt{2e}\, K_0(\frac{1}{2})$, $B = \sqrt{2e}\, K_1(\frac{1}{2})$, and we do not know why. This note assembles, with complete definitions, the geometric machinery that we conjecture explains it: the twisted de Rham cohomology of the multiplicative group $\mathbb{G}_m$ with superpotential $f(u) = \frac{1}{4}(u + u^{-1})$; its rapid-decay homology, whose two integral generators pair with cohomology to give exactly $K_0(\frac{1}{2})$, $K_1(\frac{1}{2})$ (and $I_0(\frac{1}{2})$, $I_1(\frac{1}{2})$), with period determinant $-8\pi i$; the small quantum cohomology ring of $\mathbb{P}^1$, whose quantum differential equation is the modified Bessel equation; Givental's mirror theorem identifying the two, forcing $q = \frac{1}{16}$; and Iritani's $\widehat{\Gamma}$-integral structure, which explains in general why Γ-kernel period matrices are integral over K-theory lattices. We state the resulting conjecture for the matching matrices, derive three falsifiable predictions, and record honestly what is proved, what is standard, and what is still a gap.

Contents

1 Introduction: the chain of identifications **2**

2 The Landau–Ginzburg model on $\mathbb{G}_m$ **2**
2.1 The space 2
2.2 Twisted de Rham cohomology 3

3 Rapid-decay homology and the period pairing **4**
3.1 Cycles 4
3.2 The two cycles for $f = \frac{1}{4}(u + u^{-1})$ 4
3.3 The period matrix and the quadratic relation 5

4 Quantum cohomology of $\mathbb{P}^1$ **5**
4.1 Classical cohomology and stable maps 5
4.2 The quantum product 6

5 The quantum connection and the Bessel equation **7**

6 The mirror theorem **7**

7 Integral structures: K-theory, the $\widehat{\Gamma}$-class, and Dubrovin's theorem **8**

8 The Weil representation: why $\lambda = \frac{1}{4}, \frac{3}{4}$ **9**
8.1 The representation 9
8.2 Two Cartan subalgebras, two models 10
8.3 Three payoffs 11

9 The conjecture for the matching matrices **11**
9.1 Γ-kernels are Bessel periods: an identity 11
9.2 The conjecture 12
9.3 Falsifiable predictions 12
9.4 What is proved, what is standard, what is a gap 13

1 Introduction: the chain of identifications

The papers `matching-coefficients.tex` and `meixner-pollaczek2.tex` establish, numerically and in part rigorously, the following facts. The Mellin transform on the critical line $s = \frac{1}{2} + 2ix$ carries the even Hermite functions to the Meixner–Pollaczek dual functions $F_n = \Gamma(\frac{1}{4} + ix)\widehat{P}_n$ of the $\lambda = \frac{1}{4}$ family, and the odd Hermite functions to the $\lambda = \frac{3}{4}$ family. The matching coefficients

$$c_{mk} \;=\; \int_{\mathbb{R}} \widehat{P}_m\, \widehat{P}_k\, \Gamma(\tfrac{1}{4} + ix)^2\, \Gamma(\tfrac{1}{4} - ix)\, dx$$

satisfy the empirical phase law $c_{mk} = (-i)^{|m-k|}|c_{mk}|$, and after removing the phase every entry lies in the rank-two lattice

$$\mathbb{Z}A \oplus \mathbb{Z}B, \qquad A = \sqrt{2e}\, K_0(\tfrac{1}{2}), \qquad B = \sqrt{2e}\, K_1(\tfrac{1}{2}). \tag{1}$$

Why a *lattice*? Why *rank two*? Why *Bessel*, and why the argument $\frac{1}{2}$?

This note proposes an answer in the form of a chain of standard identifications:

$$\begin{aligned} \Gamma\text{-kernel integrals} \;&\longleftrightarrow\; \text{periods of } (\mathbb{G}_m,\, f = \tfrac{1}{4}(u + u^{-1})) \\ &\longleftrightarrow\; \text{flat sections of the quantum connection of } \mathbb{P}^1 \text{ at } q = \tfrac{1}{16} \\ &\longleftrightarrow\; \text{the K-theory lattice } K^0(\mathbb{P}^1) = \mathbb{Z}[\mathcal{O}] \oplus \mathbb{Z}[\mathcal{O}(1)] \text{ via } \widehat{\Gamma}. \end{aligned}$$

Each link in the chain is established mathematics; what is conjectural is that the matching matrices of our papers are an instance of it. Sections 2–7 define everything; Section 8 adds a second, independent roof — the Weil representation, which explains the parameters $\frac{1}{4}, \frac{3}{4}$, the role of Mellin, and the phases; the final section states the conjecture and its testable consequences.

2 The Landau–Ginzburg model on $\mathbb{G}_m$

2.1 The space

Definition 2.1 (Multiplicative group)**.** $\mathbb{G}_m$ denotes the algebraic group $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$, with coordinate ring the Laurent polynomials $\mathcal{O}(\mathbb{G}_m) = \mathbb{C}[u, u^{-1}]$ and invariant 1-form du/u. As a topological space it is a twice-punctured sphere (punctures at 0 and ∞). Its compact real form is the circle $|u| = 1$; its split real form is $\mathbb{R}_+$, the home of the Mellin transform.

Definition 2.2 (Superpotential)**.** A *Landau–Ginzburg model* is a pair (X, f) of a smooth affine variety X and a regular function $f\colon X \to \mathbb{C}$, called the *superpotential*. A *critical point* of f is a point where $df = 0$; f is a *Morse* superpotential if all critical points are nondegenerate (Hessian invertible) with distinct critical values.

Our object of study is

$$X = \mathbb{G}_m, \qquad f(u) = \tfrac{1}{4}\big(u + u^{-1}\big). \tag{2}$$

Then $df = \frac{1}{4}(1-u^{-2})\,du$ vanishes at $u = \pm 1$, with critical values $f(\pm 1) = \pm\frac{1}{2}$: two Morse critical points. The map $f\colon \mathbb{G}_m \to \mathbb{C}$ is a double cover branched over $\pm\frac{1}{2}$, with deck transformation $u \mapsto u^{-1}$; this involution fixes f, swaps the punctures $0 \leftrightarrow \infty$, and fixes both critical points.

2.2 Twisted de Rham cohomology

Definition 2.3 (Twisted de Rham complex)**.** Let (X, f) be a Landau–Ginzburg model with X an affine curve. The *twisted de Rham complex* is

$$0 \longrightarrow \mathcal{O}(X) \xrightarrow{\nabla_f} \Omega^1(X) \longrightarrow 0, \qquad \nabla_f(g) = dg - g\,df,$$

i.e. the de Rham complex of the trivial line bundle with the connection $\nabla_f = d - df\wedge$. Formally $\nabla_f = e^f \circ d \circ e^{-f}$: a 1-form is ∇_f-exact iff it becomes exact after multiplication by e^{-f}. The *twisted de Rham cohomology* is

$$H^0_{\mathrm{dR}}(X, f) = \ker \nabla_f, \qquad H^1_{\mathrm{dR}}(X, f) = \Omega^1(X) \,/\, \nabla_f\, \mathcal{O}(X).$$

Remark 2.4. The connection ∇_f has *irregular* singularities at the punctures wherever f has a pole. For (2), f has a simple pole at both $u = 0$ and $u = \infty$, so both punctures are irregular of slope 1. This is what removes the periods from the classical (regular-singular, hypergeometric) world and places them in the exponential (Bessel, Kloosterman) world.

Proposition 2.5 (Computation of the cohomology)**.** *For (X, f) as in (2):*

$$H^0_{\mathrm{dR}} = 0, \qquad H^1_{\mathrm{dR}} = \mathbb{C}\left[\tfrac{du}{u}\right] \oplus \mathbb{C}\left[u\,\tfrac{du}{u}\right],$$

and in H^1_{dR} the classes $\omega_\nu := u^\nu\,du/u$ ($\nu \in \mathbb{Z}$) satisfy the three-term relation

$$\omega_{\nu+1} \;=\; 4\nu\,\omega_\nu + \omega_{\nu-1}. \tag{3}$$

Proof. H^0: $\nabla_f g = 0$ means $g = c\,e^f$, which is not a Laurent polynomial unless $c = 0$.

H^1: applying ∇_f to the monomial u^ν,

$$\nabla_f(u^\nu) = \nu u^\nu\,\tfrac{du}{u} - \tfrac{1}{4}\big(u^{\nu+1} - u^{\nu-1}\big)\tfrac{du}{u} = \tfrac{1}{4}\big(4\nu\,\omega_\nu - \omega_{\nu+1} + \omega_{\nu-1}\big),$$

so in the quotient $\omega_{\nu+1} = 4\nu\,\omega_\nu + \omega_{\nu-1}$, which is (3). Reading it upward reduces every ω_ν with $\nu \geq 2$ to ω_0, ω_1; reading it downward ($\omega_{\nu-1} = \omega_{\nu+1} - 4\nu\,\omega_\nu$) does the same for $\nu \leq -1$. Hence ω_0, ω_1 span. They are independent because the period pairing of Section 3 is nondegenerate on them (Theorem 3.6: the period determinant is $-8\pi i \neq 0$). □

Proposition 2.6 (Dimension = number of critical points)**.** *For a Morse superpotential f on an affine curve X, $\dim H^1_{\mathrm{dR}}(X, f)$ equals the number of critical points of f.*

This is the algebraic avatar of Morse theory (for the exponential setting see e.g. the framework of Katz, or Kouchnirenko's theorem in the Newton-polytope form: for a Laurent polynomial in one variable with Newton segment $[a, b]$ the dimension is $b - a$; here $[-1, 1]$ gives 2). It is the structural source of the ranks in our papers:

weight	space	superpotential	crit. pts	rank
Gaussian $e^{-t^2/2}$	$\mathbb{A}^1$	$t^2/2$	1 ($t = 0$)	1
Bessel kernel	$\mathbb{G}_m$	$\frac{1}{4}(u + u^{-1})$	2 ($u = \pm 1$)	2

One critical point $\Rightarrow$ the Hermite matching matrix of `hermite-matching-matrix.tex` is integral over a *single* transcendental constant. Two critical points $\Rightarrow$ the rank-two lattice (1).

3 Rapid-decay homology and the period pairing

3.1 Cycles

To integrate $e^{-f}\omega$ we need chains on which e^{-f} decays, or which have no boundary at all.

Definition 3.1 (Rapid-decay homology, after Bloch–Esnault)**.** Let (X, f) be as above, X a curve with punctures. A *rapid-decay chain* is a (possibly unbounded) topological chain in X whose ends approach the punctures only within sectors where $\operatorname{Re} f \to +\infty$ (so that e^{-f} decays faster than any power). The *rapid-decay homology* $H_1^{\mathrm{rd}}(X, f)$ is the homology of the complex of such chains; it carries a natural $\mathbb{Z}$-structure $H_1^{\mathrm{rd}}(X, f; \mathbb{Z})$, the classes of integral chains. For Morse f one basis is given by the *Lefschetz thimbles*: the steepest-descent paths of $\operatorname{Re} f$ through the critical points.

Definition 3.2 (Period pairing)**.** The *period pairing* is

$$H_1^{\mathrm{rd}}(X, f) \times H^1_{\mathrm{dR}}(X, f) \longrightarrow \mathbb{C}, \qquad \langle \gamma, \omega \rangle = \int_\gamma e^{-f}\, \omega.$$

It is well defined (∇_f-exact forms integrate to zero because $e^{-f}\nabla_f g = d(e^{-f} g)$ and the boundary terms die in the rapid-decay sectors) and perfect (Bloch–Esnault). Its values are called *exponential periods*.

3.2 The two cycles for $f = \frac{1}{4}(u + u^{-1})$

Near $u = \infty$, $f \sim u/4$, so the rapid-decay sector is $\operatorname{Re} u > 0$; near $u = 0$, $f \sim 1/(4u)$, rapid decay again requires $\operatorname{Re} u > 0$. Two independent integral classes:

- γ_1: *the positive real axis* $\mathbb{R}_+$, oriented $0 \to \infty$. On $\mathbb{R}_+$, f is real, tends to $+\infty$ at both ends, and has its minimum $\frac{1}{2}$ at $u = 1$: so $\mathbb{R}_+$ is precisely the Lefschetz thimble through the critical point $u = +1$. *The Mellin half-line is itself a homology class.*
- γ_2: *the unit circle* $|u| = 1$, counterclockwise. A closed loop needs no decay condition. (The thimble through $u = -1$ is an integral combination of γ_1 and γ_2; the circle is the more convenient generator.)

Proposition 3.3 (The periods are Bessel values)**.** *With $\omega_\nu = u^\nu\,du/u$ and $\nu \in \mathbb{Z}$:*

$$\langle \gamma_1, \omega_\nu \rangle = \int_0^\infty e^{-\frac{1}{4}(u+u^{-1})}\, u^\nu\, \frac{du}{u} = 2\,K_\nu(\tfrac{1}{2}), \tag{4}$$

$$\langle \gamma_2, \omega_\nu \rangle = \oint_{|u|=1} e^{-\frac{1}{4}(u+u^{-1})}\, u^\nu\, \frac{du}{u} = 2\pi i\,(-1)^\nu\, I_\nu(\tfrac{1}{2}). \tag{5}$$

Proof. (4) is the classical integral representation $\int_0^\infty e^{-\frac{z}{2}(u+u^{-1})} u^\nu\, du/u = 2K_\nu(z)$ at $z = \frac{1}{2}$. (5): the generating function $e^{\frac{z}{2}(u+u^{-1})} = \sum_{\mu\in\mathbb{Z}} I_\mu(z)\,u^\mu$ at $z = -\frac{1}{2}$ gives $e^{-\frac{1}{4}(u+u^{-1})} = \sum_\mu (-1)^\mu I_\mu(\frac{1}{2})\,u^\mu$ (using $I_\mu(-z) = (-1)^\mu I_\mu(z)$ for integer μ), and the contour integral extracts the coefficient of $u^{-\nu}$, which is $(-1)^\nu I_\nu(\frac{1}{2})$ since $I_{-\nu} = I_\nu$. □

Remark 3.4 (The cohomology relation is the Bessel recurrence). Pairing relation (3) against γ_1 and γ_2 must give identities, and does: $K_{\nu+1}(z) = \frac{2\nu}{z}K_\nu(z) + K_{\nu-1}(z)$ at $z = \frac{1}{2}$ reads $K_{\nu+1} = 4\nu K_\nu + K_{\nu-1}$, and $a_\nu := (-1)^\nu I_\nu(\frac{1}{2})$ satisfies $a_{\nu+1} = 4\nu a_\nu + a_{\nu-1}$ by the I-Bessel recurrence. The three-term recurrence of the Bessel family *is* the relation $\nabla_f(u^\nu) \equiv 0$ in cohomology.

3.3 The period matrix and the quadratic relation

Definition 3.5 (Period matrix)**.** Relative to the bases (γ_1, γ_2) of $H_1^{\mathrm{rd}}(\mathbb{Z})$ and (ω_0, ω_1) of H^1_{dR}, the *period matrix* is

$$P \;=\; \begin{pmatrix} 2K_0(\frac{1}{2}) & 2K_1(\frac{1}{2}) \\ 2\pi i\, I_0(\frac{1}{2}) & -2\pi i\, I_1(\frac{1}{2}) \end{pmatrix}.$$

Theorem 3.6 (Legendre-type period relation)**.** $\det P = -8\pi i$.

Proof. $\det P = -4\pi i\,\big(K_0I_1 + K_1I_0\big)(\frac{1}{2})$. The Wronskian identity for the modified Bessel equation is $I_0(z)K_1(z) + I_1(z)K_0(z) = 1/z$; at $z = \frac{1}{2}$ the bracket is 2. □

This is the exponential-period analogue of the Legendre relation $\eta_1\omega_2 - \eta_2\omega_1 = 2\pi i$ for elliptic periods: Poincaré duality forces the determinant of a full period matrix to be an explicit power of $2\pi i$ times a rational number. The transcendentals K_0, K_1, I_0, I_1 at $\frac{1}{2}$ are constrained by exactly one algebraic relation, and this is it. (This is the mechanism behind the quadratic relations among Bessel moments of Broadhurst, proved by Fresán–Sabbah–Yu.)

4 Quantum cohomology of $\mathbb{P}^1$

We now build the B-side[1] partner of Sections 2–3 from scratch.

4.1 Classical cohomology and stable maps

Definition 4.1. $H^*(\mathbb{P}^1;\mathbb{C}) = \mathbb{C}\,1 \oplus \mathbb{C}\,h$, where 1 is the fundamental class (degree 0) and h is the point class (degree 2), with $h^2 = 0$ and Poincaré pairing $g(1,h) = \int_{\mathbb{P}^1} h = 1$, $g(1,1) = g(h,h) = 0$. The dual basis of $(1,h)$ under g is $(h,1)$.

[1] Conventions differ on which side is "A" and which "B"; we call the curve-counting side A and the Landau–Ginzburg side B and will not fuss about it.

Definition 4.2 (Stable maps and Gromov–Witten invariants)**.** For $d \geq 0$ let $\overline{M}_{0,n}(\mathbb{P}^1, d)$ be the moduli space of *stable maps*: degree-d holomorphic maps $\varphi\colon C \to \mathbb{P}^1$ from a connected nodal genus-0 curve with n marked smooth points, such that φ has finitely many automorphisms fixing the marks. It is a compact orbifold of complex dimension

$$\dim \overline{M}_{0,n}(\mathbb{P}^1, d) = \dim \mathbb{P}^1 - 3 + \textstyle\int_d c_1(T\mathbb{P}^1) + n = 2d + n - 2,$$

using $c_1(T\mathbb{P}^1) = 2h$. With evaluation maps $\mathrm{ev}_i\colon \overline{M}_{0,n}(\mathbb{P}^1, d) \to \mathbb{P}^1$, $\mathrm{ev}_i(\varphi) = \varphi(p_i)$, the *genus-zero Gromov–Witten invariants* are

$$\langle \gamma_1, \dots, \gamma_n \rangle_d = \int_{\overline{M}_{0,n}(\mathbb{P}^1, d)} \mathrm{ev}_1^*\gamma_1 \cup \cdots \cup \mathrm{ev}_n^*\gamma_n,$$

nonzero only when the total codimension of the γ_i equals $2d + n - 2$. They are (virtual) counts of rational curves meeting the cycles Poincaré-dual to the γ_i.

Lemma 4.3. *$\langle h, h, h \rangle_1 = 1$, and this is the only nonzero 3-point invariant of positive degree with all insertions from $\{1, h\}$.*

Proof. Dimension count: $n = 3$ needs total codimension $2d + 1$; insertions from $\{1, h\}$ have codimension ≤ 3, forcing $d = 1$ and all three insertions equal to h. The invariant counts degree-1 maps $\mathbb{P}^1 \to \mathbb{P}^1$ (i.e. Möbius transformations) sending three marked points to three specified points; by sharp 3-transitivity of PGL_2 there is exactly one. □

4.2 The quantum product

Definition 4.4 (Small quantum product)**.** Introduce a formal variable q (the *Novikov variable*) recording the degree. The *small quantum product* on $H^*(\mathbb{P}^1) \otimes \mathbb{C}[q]$ is defined by

$$g(\alpha * \beta,\, \gamma) \;=\; \sum_{d \geq 0} q^d \, \langle \alpha, \beta, \gamma \rangle_d,$$

i.e. the classical cup product corrected by curve counts. It is associative (a nontrivial theorem, the WDVV equations) and graded if q is given degree $2\textstyle\int_d c_1 = 4$.

Theorem 4.5. *$QH^*(\mathbb{P}^1) = \mathbb{C}[h, q] \,/\, (h * h - q)$. In particular $h * h = q \cdot 1$: "the square of a point is a curve."*

Proof. $g(h * h, \gamma) = \langle h, h, \gamma \rangle_0 + q\, \langle h, h, \gamma \rangle_1 + \cdots$. The $d = 0$ invariants are classical: $\langle h, h, \gamma \rangle_0 = \int h \cup h \cup \gamma = 0$. By Lemma 4.3 the only quantum term is $\gamma = h$, $d = 1$, giving $g(h * h, h) = q$, and $g(h * h, 1) = 0$. Since $g(1, h) = 1$, this says $h * h = q \cdot 1$. □

Proposition 4.6 (Semisimplicity: the two vacua)**.** *For $q \neq 0$ the algebra $QH^*(\mathbb{P}^1)_q = \mathbb{C}[h]/(h^2 - q)$ is semisimple, with idempotents $e_\pm = \frac{1}{2}\big(1 \pm h/\sqrt{q}\big)$ and with quantum multiplication by h having eigenvalues $\pm\sqrt{q}$.*

The two idempotents are the algebraic shadow of the two critical points of Section 2; the precise match is Theorem 6.2.

5 The quantum connection and the Bessel equation

Definition 5.1 (Quantum connection)**.** Fix $\hbar \in \mathbb{C}^\times$ and set $q = e^t$. The *quantum connection* (Dubrovin/Givental connection) on the trivial bundle with fibre $H^*(\mathbb{P}^1)$ over the q-line is

$$\nabla_{q\partial_q} = \hbar\, q\frac{\partial}{\partial q} \;-\; (h*).$$

A *flat section* is $S(q)$ with $\nabla_{q\partial_q} S = 0$.

Proposition 5.2 (QDE = modified Bessel)**.** *Write $S = s_0(q)\,1 + s_1(q)\,h$. Flatness is the system $\hbar\, q s_0' = q\, s_1$, $\hbar\, q s_1' = s_0$ (using $h*1 = h$, $h*h = q$), hence*

$$\big(\hbar\, q\tfrac{d}{dq}\big)^2 s_0 \;=\; q\, s_0 :$$

the modified Bessel equation. Its solutions at $\hbar = 1$ are spanned by $I_0(2\sqrt{q}\,)$ and $K_0(2\sqrt{q}\,)$.

Proof. Substitute $z = 2\sqrt{q}/\hbar$: $(q\,d/dq)^2 = \frac{1}{4}(z\,d/dz)^2$ and the equation becomes $z^2 w'' + zw' - z^2 w = 0$, the modified Bessel equation of order 0. □

Definition 5.3 (J-function)**.** The *J-function* of $\mathbb{P}^1$ is the H^*-valued flat generating series of gravitational descendants; for $\mathbb{P}^1$ it has the closed form (Givental)

$$J_{\mathbb{P}^1}(q,\hbar) = \sum_{d\ge 0} \frac{q^d}{\prod_{r=1}^{d}(h + r\hbar)^2} \;\overset{h^2=0}{=}\; \sum_{d\ge 0} \frac{q^d}{(d!)^2\hbar^{2d}}\Big(1 - \tfrac{2h}{\hbar}\,H_d\Big) + \cdots,$$

whose degree-0 component at $\hbar = 1$ is $\sum_d q^d/(d!)^2 = I_0(2\sqrt{q}\,)$, with H_d the harmonic numbers appearing in the h-component.

Thus *I-Bessel is the curve-counting function of $\mathbb{P}^1$*: the coefficient of q^d is $1/(d!)^2$, the reciprocal of the automorphism-weighted count of degree-d covers with the requisite markings. K-Bessel is the transcendental second solution — the flat section singled out not by holomorphy at $q = 0$ but (as we recall in Section 7) by the integral structure.

6 The mirror theorem

Definition 6.1 (Oscillatory integrals)**.** For the Landau–Ginzburg model $(\mathbb{G}_m, W_q)$ with $W_q(u) = u + q/u$ and a class $\gamma \in H_1^{\mathrm{rd}}(\mathbb{G}_m, W_q/\hbar)$, the *oscillatory integral* is

$$\mathcal{I}_\gamma(q) = \int_\gamma e^{-W_q(u)/\hbar}\,\frac{du}{u}.$$

Theorem 6.2 (Givental's mirror theorem for $\mathbb{P}^1$)**.** *The oscillatory integrals $\mathcal{I}_\gamma$ are exactly the solutions of the quantum differential equation of $\mathbb{P}^1$ (Proposition 5.2), and the assignment $u^\nu\, du/u \mapsto$ (class in $H^*(\mathbb{P}^1)$) identifies*

$$H^1_{\mathrm{dR}}\big(\mathbb{G}_m,\, W_q/\hbar\big) \;\cong\; QH^*(\mathbb{P}^1)_q$$

compatibly with connections. Under this identification, quantum multiplication by h corresponds to multiplication by u in the Jacobi ring $\mathbb{C}[u,u^{-1}]/(W_q')$, and the eigenvalues $\pm 2\sqrt{q}$ of the grading operator match the critical values of W_q.

We do not reprove Givental's theorem, but the differential equation is a two-line check worth recording.

Proposition 6.3. *At $\hbar = 1$ let $m_\nu(q) = \int_\gamma e^{-W_q} u^\nu \, du/u$. Then $\big(q\frac{d}{dq}\big)^2 m_0 = q\, m_0$.*

Proof. Exactness of $d\big(e^{-W_q} u^\nu\big)$ on γ gives the moment recursion $m_{\nu+1} = \nu\, m_\nu + q\, m_{\nu-1}$. Since $\partial_q e^{-W_q} = -(1/u)e^{-W_q}$ we get $q\,\partial_q m_\nu = -q\, m_{\nu-1}$, hence $(q\partial_q)^2 m_0 = -q\, m_{-1} + q^2 m_{-2}$, and the recursion at $\nu = -1$ ($m_0 = -m_{-1} + q\, m_{-2}$) turns this into $q\, m_0$. □

Corollary 6.4 (Why $q = \frac{1}{16}$)**.** *The substitution $u = 4u'$ carries our superpotential* (2) *to the mirror superpotential:*

$$\tfrac{1}{4}\big(u + u^{-1}\big) = u' + \frac{1/16}{u'} = W_{1/16}(u').$$

Equivalently: the critical values $\pm\frac{1}{2}$ of f must equal the eigenvalues $\pm 2\sqrt{q}$ of quantum multiplication, forcing $q = \frac{1}{16}$; and the Bessel argument $2\sqrt{q} = \frac{1}{2}$ in Proposition 3.3 is the same statement. The argument $\frac{1}{2}$ in $K_0(\frac{1}{2}), K_1(\frac{1}{2})$ is not fitted: it is the critical value of the superpotential, i.e. the point on the q-line where the matching-coefficient geometry sits.

So the constants (1) are, up to the normalization $\sqrt{2e}$, values of the flat sections of the quantum connection of $\mathbb{P}^1$ at $q = \frac{1}{16}$, $\hbar = 1$.

7 Integral structures: K-theory, the $\widehat{\Gamma}$-class, and Dubrovin's theorem

The period pairing of Section 3 lands in $\mathbb{C}$; the lattice (1) demands an integral structure. On the homology side it is $H_1^{\mathrm{rd}}(\mathbb{Z})$; the modern formulation on the quantum side, due to Iritani and Katzarkov–Kontsevich–Pantev, is K-theoretic.

Definition 7.1 (K-theory of $\mathbb{P}^1$)**.** $K^0(\mathbb{P}^1)$ is the Grothendieck group of algebraic vector bundles on $\mathbb{P}^1$: formal differences $[E] - [F]$ modulo $[E] = [E'] + [E'']$ for every short exact sequence. It is free of rank 2:

$$K^0(\mathbb{P}^1) = \mathbb{Z}[\mathcal{O}] \oplus \mathbb{Z}[\mathcal{O}(1)],$$

where $\mathcal{O}(n)$ is the line bundle of degree n. The *Chern character* $\mathrm{ch}\colon K^0(\mathbb{P}^1) \to H^*(\mathbb{P}^1;\mathbb{Q})$ is the ring map $\mathrm{ch}(\mathcal{O}(n)) = e^{nh} = 1 + nh$ (as $h^2 = 0$).

Definition 7.2 (Euler pairing)**.** $\chi(E,F) = \sum_i (-1)^i \dim \mathrm{Ext}^i(E,F)$. On $\mathbb{P}^1$, in the basis $(\mathcal{O}, \mathcal{O}(1))$:

$$\chi = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

since $\chi(\mathcal{O}, \mathcal{O}(1)) = \dim H^0(\mathbb{P}^1, \mathcal{O}(1)) = 2$ (the two sections x, y) and $\chi(\mathcal{O}(1), \mathcal{O}) = \chi(\mathcal{O}(-1)) = 0$.

Definition 7.3 ($\widehat{\Gamma}$-class)**.** For a vector bundle T with Chern roots δ_i, set $\widehat{\Gamma}(T) = \prod_i \Gamma(1 + \delta_i)$, expanded by

$$\log \Gamma(1+x) = -\gamma x + \sum_{k \geq 2} \frac{(-1)^k \zeta(k)}{k}\, x^k.$$

For $\mathbb{P}^1$ (Euler sequence: $T\mathbb{P}^1$ has the roots of $\mathcal{O}(1)^{\oplus 2}$ modulo a trivial factor):

$$\widehat{\Gamma}_{\mathbb{P}^1} = \Gamma(1+h)^2 = 1 - 2\gamma\, h \pmod{h^2}.$$

Definition 7.4 ($\widehat{\Gamma}$-integral structure, Iritani)**.** To a class $E \in K^0(\mathbb{P}^1)$ associate the flat section of the quantum connection whose asymptotic class at $q \to 0$ is

$$s(E) \;=\; \frac{1}{(2\pi)^{\dim/2}}\,\widehat{\Gamma}_{\mathbb{P}^1} \cup (2\pi i)^{\deg/2}\,\mathrm{ch}(E)$$

(the operator $(2\pi i)^{\deg/2}$ multiplies H^{2k} by $(2\pi i)^k$). The $\widehat{\Gamma}$*-integral structure* is the $\mathbb{Z}$-local system of flat sections $\{s(E) : E \in K^0(\mathbb{P}^1)\}$.

Theorem 7.5 (Iritani; Katzarkov–Kontsevich–Pantev)**.** *The $\widehat{\Gamma}$-integral structure is preserved by the monodromy and Stokes automorphisms of the quantum connection, and under the mirror identification (Theorem 6.2) it corresponds to the integral rapid-decay homology $H_1^{\mathrm{rd}}(\mathbb{G}_m, W_q; \mathbb{Z})$ of Section 3. Moreover the pairing of flat sections matches the Euler pairing χ on K^0.*

This theorem is the general answer to "why do Γ-kernel integrals produce integer matrices over transcendental lattices": *the Γ-factors are the $\widehat{\Gamma}$-class, the mechanism that rotates the topological $\mathbb{Z}$-lattice into the algebraic de Rham frame.* Whenever one expands periods of an exponential connection in an algebraic basis, the coefficients lie in the $\mathbb{Z}$-span of finitely many periods — one for each generator of the K-theory (equivalently, rapid-decay homology) lattice.

Definition 7.6 (Exceptional collection; Stokes matrix)**.** An *exceptional collection* in the derived category $D^b(\mathbb{P}^1)$ is a sequence $(E_1, \dots, E_r)$ with $\mathrm{Ext}^\bullet(E_i, E_i) = \mathbb{C}$ and $\mathrm{Ext}^\bullet(E_j, E_i) = 0$ for $j > i$, generating D^b. For $\mathbb{P}^1$: $(\mathcal{O}, \mathcal{O}(1))$ (Beilinson). The *Stokes matrix* of the quantum connection is the unipotent matrix comparing the two sectorial bases of flat sections at the irregular point $q = \infty$ along the two Stokes directions.

Theorem 7.7 (Dubrovin's conjecture for $\mathbb{P}^1$; proved by Dubrovin, Guzzetti)**.** *The Stokes matrix of the quantum connection of $\mathbb{P}^1$ is the Euler matrix of the exceptional collection:*

$$S = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad S_{12} = \chi(\mathcal{O}, \mathcal{O}(1)) = 2.$$

Remark 7.8 (The two 2's). The off-diagonal Stokes entry $2 = \dim H^0(\mathbb{P}^1, \mathcal{O}(1))$ and the Wronskian value $\big(I_0K_1 + I_1K_0\big)(\frac{1}{2}) = 2$ of Theorem 3.6 are cousins, not strangers: both are computed from the same rank-two irregular connection, one from its Stokes data, one from its period determinant. We flag, without claiming, that a uniform statement should exist.

8 The Weil representation: why $\lambda = \frac{1}{4}, \frac{3}{4}$

The geometry of Sections 2–7 explains the *lattice*. A parallel and equally classical structure explains the *cast of characters* — why Hermite, why Meixner–Pollaczek, why the parameters $\frac{1}{4}$ and $\frac{3}{4}$, and why the Mellin transform intertwines them. The statement (Robin's observation): *the Mellin-sends-Hermite-to-MP theorem is the explicit unitary intertwiner between the Schrödinger model and the Mellin model of the Weil representation.*

8.1 The representation

Definition 8.1 (Heisenberg group; Stone–von Neumann)**.** The Heisenberg group H_1 is $\mathbb{R}^2 \times \mathbb{R}$ with $(p, q, z)(p', q', z') = (p{+}p',\ q{+}q',\ z{+}z'{+}\frac{1}{2}(pq'{-}qp'))$. By the Stone–von Neumann theorem it

has, for each nontrivial central character, a *unique* irreducible unitary representation: on $L^2(\mathbb{R})$, with q acting by translation and p by multiplication by $e^{2\pi it}$-type characters (the *Schrödinger model*).

Definition 8.2 (Metaplectic group and Weil representation)**.** $SL_2(\mathbb{R})$ acts on H_1 by automorphisms fixing the centre; uniqueness in Stone–von Neumann then gives a projective representation of $SL_2(\mathbb{R})$ on $L^2(\mathbb{R})$, which lifts to an honest unitary representation ω of the double cover $Mp_2(\mathbb{R})$: the *Weil* (oscillator, metaplectic) representation. On the Lie algebra $\mathfrak{sl}_2 = \langle e, h, f\rangle$ it acts by

$$\omega(e) = \tfrac{i}{2}\, t^2, \qquad \omega(f) = \tfrac{i}{2}\, \frac{d^2}{dt^2}, \qquad \omega(h) = t\frac{d}{dt} + \tfrac{1}{2}.$$

Proposition 8.3 (Irreducible decomposition)**.** *$\omega = \omega^+ \oplus \omega^-$, the even and odd subspaces of $L^2(\mathbb{R})$. These are irreducible lowest-weight representations of $Mp_2(\mathbb{R})$ with lowest K-weights*

$$\tfrac{1}{2} = 2 \cdot \tfrac{1}{4} \qquad \textit{and} \qquad \tfrac{3}{2} = 2 \cdot \tfrac{3}{4}.$$

In the normalization $D(\lambda)$ (weight $= 2\lambda$) these are exactly the Meixner–Pollaczek parameters: the two MP families are the two irreducible components of the Weil representation. The even/odd split of the Hermite family (§8 of `matching-coefficients.tex`*: "one family on $\mathbb{R}$, two on $\mathbb{R}_+$") is the irreducible decomposition.*

8.2 Two Cartan subalgebras, two models

Inside $\mathfrak{sl}_2$ there are two conjugacy classes of Cartan subalgebra, and each gives a "model" of ω by diagonalization:

	elliptic (compact $K = SO(2)$)	hyperbolic (split torus A)
generator via ω	$\frac{1}{2}\big(-\frac{d^2}{dt^2} + t^2\big)$ (oscillator)	$\frac{1}{2i}\big(t\frac{d}{dt} + \frac{1}{2}\big)$ (dilation)
spectrum	discrete, $n + \frac{1}{2}$	continuous, $x \in \mathbb{R}$
eigenbasis	Hermite ψ_n	$\lvert t\rvert^{-1/2+2ix}$ (generalized)
model	Schrödinger	Mellin

Theorem 8.4 (The intertwiner)**.** *The unitary map diagonalizing the dilation generator is the Mellin transform on the critical line $s = \frac{1}{2} + 2ix$ (restricted to even or odd functions, i.e. to $\omega^\pm$). It intertwines the Schrödinger model with the Mellin model, and its matrix elements against the K-basis — the overlaps $\langle\, |t|^{-1/2+2ix}\,, \psi_n\,\rangle$ — are, up to normalization, the Meixner–Pollaczek dual functions*

$$F_n(x) = \Gamma(\lambda + ix)\, \widehat{P}_n^{(\lambda)}(x), \qquad \lambda = \tfrac{1}{4}\ (n \textit{ even}), \quad \lambda = \tfrac{3}{4}\ (n \textit{ odd}),$$

with $|\Gamma(\lambda + ix)|^2$ appearing as the Plancherel density of the hyperbolic basis. This is the content of the Mellin→MP theorem of `matching-coefficients.tex` *§8, restated representation-theoretically.*

The overlap coefficients between elliptic and hyperbolic bases of $SL_2(\mathbb{R})$ representations are classical (Basu–Wolf's subgroup-reduction tables; Koornwinder; Vilenkin–Klimyk), but there they are an entry in a table — the matching-matrix arithmetic built on top of them appears to be new.

8.3 Three payoffs

1. **The phase law.** K acts on ω by $e^{-i\theta(N+1/2)}$ (N the number operator); the Fourier transform is rotation by $\theta = \frac{\pi}{2}$ up to a metaplectic phase in μ_8 — the same eighth roots of unity that govern the phases of the matching matrices. The unproved phase law $c_{mk} = (-i)^{|m-k|}|c_{mk}|$ should follow by writing the matching integral as a matrix coefficient of an explicit group element, with the phase supplied by the K-action: a proof strategy, not merely an explanation.

2. **The order–argument gap (§9.4, item 1).** Proposition 9.1 put the spectral variable in the Bessel *order*; the passage to integer order and fixed argument is the Kontorovich–Lebedev transform — which is precisely A-side Plancherel harmonic analysis on $SL_2(\mathbb{R})$. The main technical gap of this paper is a known transform on the group.

3. **Tate's thesis.** The pairing $\langle F_m, G_n\rangle$ with kernel $\Gamma(\frac{1}{4}+ix)\Gamma(\frac{1}{4}-ix) = |\Gamma(s/2)|^2$ is the archimedean local zeta-integral: $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ is the local L-factor of $\mathbb{Q}$ at the real place, and the involution $F \leftrightarrow G$ under $s \mapsto 1-s$ is the local functional equation with its ε-factor. The zeta connection is not an appendix: the entire matching programme is local harmonic analysis at the real place, and ζ enters only when the global sum over $\mathbb{Z}$ (the θ-function) is taken.

Remark 8.5 (The open seam). The paper now has two independent classical roofs over the same computation: the Weil representation (this section: why $\frac{1}{4}, \frac{3}{4}$, why Mellin, the phases) and the quantum cohomology of $\mathbb{P}^1$ (Sections 4–7: why rank two, why $K_\nu(\frac{1}{2})$, the lattice). What is *not* in the literature, to our knowledge, is the dictionary between them — what the $\mathbb{P}^1$-geometry means on the group side. The obvious first entry: the two saddles $u = \pm 1$ of the superpotential against the two components $\omega^\pm$ of the Weil representation. Preserved as an open disagreement between two true descriptions.

9 The conjecture for the matching matrices

9.1 Γ-kernels are Bessel periods: an identity

The bridge from the Meixner–Pollaczek weight to the geometry is already visible at the level of a Fubini exercise.

Proposition 9.1. *For* $\operatorname{Re} a, \operatorname{Re} b > 0$,

$$\Gamma(a)\,\Gamma(b) = 4\int_0^\infty t^{\,a+b}\,K_{a-b}(2t)\,\frac{dt}{t}.$$

In particular the Meixner–Pollaczek kernel is

$$\Gamma(\tfrac{1}{4}+ix)\,\Gamma(\tfrac{1}{4}-ix) = 4\int_0^\infty t^{1/2}\,K_{2ix}(2t)\,\frac{dt}{t}\,:$$

a Mellin transform (in the argument) of the K-Bessel kernel, with the spectral variable x sitting in the order.

Proof. Write $\Gamma(a)\Gamma(b) = \iint_{(0,\infty)^2} e^{-u-v}u^a v^b\,\frac{du\,dv}{uv}$ and substitute $u = tr$, $v = t/r$; then $\frac{du\,dv}{uv} = 2\,\frac{dt}{t}\frac{dr}{r}$, $u+v = t(r+r^{-1})$, $u^a v^b = t^{a+b}r^{a-b}$, and the r-integral is $2K_{a-b}(2t)$ by (4)'s general form. □

So every Γ-product weight in our papers is, literally, a period of the Bessel geometry of Sections 2–3 — but with x in the order $\nu = 2ix$, whereas the lattice (1) lives at integer orders $\nu \in \{0, 1\}$ and fixed argument $\frac{1}{2}$. The passage from continuous order to the integer-order lattice is a Kontorovich–Lebedev-type spectral decomposition, and carrying it out cleanly for the matching integrals is the main technical gap (§9.4).

9.2 The conjecture

Conjecture 9.2. *Let $C = (c_{mk})$ be the forward matching matrix of the $\lambda = \frac{1}{4}$ Meixner–Pollaczek dual pair, with phases removed as in* `meixner-pollaczek2.tex`*. Then, up to the global normalization $\sqrt{2e}$:*

(i) each entry of C is a period pairing $\langle \gamma_{mk}, \omega \rangle$ of the Landau–Ginzburg model $(\mathbb{G}_m,\ \frac{1}{4}(u + u^{-1}))$, i.e. a flat-section value of the quantum connection of $\mathbb{P}^1$ at $q = \frac{1}{16}$, with $\gamma_{mk} \in H_1^{\mathrm{rd}}(\mathbb{Z})$;

(ii) the lattice $\mathbb{Z}A \oplus \mathbb{Z}B$ is the image of the integral lattice — equivalently, via Theorem 7.5, of $K^0(\mathbb{P}^1) = \mathbb{Z}[\mathcal{O}] \oplus \mathbb{Z}[\mathcal{O}(1)]$ under the $\widehat{\Gamma}$-integral structure — in the K-directions of the period matrix P;

(iii) the integrality of C over $\mathbb{Z}A \oplus \mathbb{Z}B$ is an instance of Iritani integrality: expansion of periods in an algebraic basis has coefficients in the $\mathbb{Z}$-span of the lattice periods.

9.3 Falsifiable predictions

Prediction 9.3 (I-companions). The geometry has *four* distinguished periods, not two: K_0, K_1 (thimble $\gamma_1 = \mathbb{R}_+$) and I_0, I_1 (circle γ_2), at argument $\frac{1}{2}$. If Conjecture 9.2 holds, the constants $A' = \sqrt{2e}\,\pi I_0(\frac{1}{2})$, $B' = \sqrt{2e}\,\pi I_1(\frac{1}{2})$ (normalization to be pinned down) must appear somewhere in the theory.

Remark 9.4 (Test outcome: the naive version of Prediction 9.3 is refuted). The obvious place to look — the inverse of the forward matrix — fails, twice over, and instructively (numerics on `mp_dual_matching.pkl`, August 2026):

1. Truncated inverses C_N^{-1} *diverge* as N grows: the slow decay $|c_{0k}| \sim k^{-0.7}$ makes the truncations ill-conditioned, and entries oscillate without settling. The matrix inverse of a truncation is not the analytic inverse.

2. The *exact* analytic inverse is the biorthogonal expansion $\widehat{P}_k = \sum_m r_{mk} F_m$ with $r_{mk} = \int \widehat{P}_m \widehat{P}_k\, \Gamma(\frac{1}{4} - ix)\, dx$ — single-Γ integrals. These are *rational*: from the inverse Mellin transform $\frac{1}{2\pi}\int_{\mathbb{R}} \Gamma(\frac{1}{4} + ix)\, t^{-ix}\, dx = t^{1/4}e^{-t}$, applying $\theta = t\,d/dt$ repeatedly and setting $t = 1$,
$$\int_{\mathbb{R}} x^j\, \Gamma(\tfrac{1}{4} - ix)\, dx \;=\; i^j\, \frac{2\pi}{e}\, \rho_j, \qquad \rho_j \in \mathbb{Q},$$
$\rho_{0..4} = 1,\ \frac{3}{4},\ -\frac{7}{16},\ -\frac{53}{64},\ \frac{497}{256}$ (PSLQ-confirmed at 25+ digits, and provable by the θ-recursion $p_{j+1} = \theta p_j + (\frac{1}{4} - t)p_j$ on p_j defined by $\theta^j(t^{1/4}e^{-t}) = p_j(t)\, t^{1/4}e^{-t}$). So the reverse matrix lies in $\frac{2\pi}{e}\,\mathbb{Q}$: no Bessel values at all.

Two salvage observations. First, the reverse constant is $2\pi e^{-1} = 2\pi\, e^{-(f(1)-f(-1))}$: the exponential of (minus) the *difference of the two critical values* — in Stokes language, the instanton factor between the two vacua — whereas the forward constants carry $e^{-f(\pm 1)}$ separately. We flag this as suggestive, not established. Second, the Hermite precedent locates the I-family elsewhere: in `hermite-matching-matrix.tex` the I-Bessel appears in the *generating function of the integer matrix* (diagonal EGF $e^x I_0(2x)$), not in any inverse. The revised prediction: $I_\nu(\frac{1}{2})$-values (or the I-series termwise) appear in generating functions of the integer pair $(\mathfrak{a}_{mk}, \mathfrak{b}_{mk})$, the arithmetic shadow of the circle cycle γ_2.

Prediction 9.5 (Determinant rationality). Products pairing K-rows against I-rows must rationalize by Theorem 3.6: any 2×2 block of the full period structure built from (A, B) and (A', B') has determinant in $2\pi i\, e\, \mathbb{Q}$ (the e from $\sqrt{2e}^2$; note $AB' + A'B = \sqrt{2e}^2\, \pi(K_0 I_1 + I_0 K_1)(\frac{1}{2}) = 4\pi e$). Numerically checkable to hundreds of digits.

Prediction 9.6 (K-theory basis). There is a basis change $(A, B) \to (A'', B'')$ with $\mathrm{GL}_2(\mathbb{Z})$-matrix determined by $\mathrm{ch}(\mathcal{O}), \mathrm{ch}(\mathcal{O}(1))$ and the $\widehat{\Gamma}$-class, in which the matching matrix becomes distinguished — e.g. triangular against the Euler form $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, or with unimodular leading blocks. If no $\mathrm{GL}_2(\mathbb{Z})$ change of basis makes the matrix structurally nicer, the K-theoretic reading is wrong even if (i) holds.

Prediction 9.7 (Rank law for other kernels). For the other weights in the programme (`orthogonal-plan.tex`) the lattice rank should be computable *before any integrals*: it is the number of critical points of the superpotential attached to the kernel by the analogue of Proposition 9.1. Gaussian: 1 (observed). Bessel/MP: 2 (observed). A kernel with r Γ-factor pairs should produce the geometry of an $(r-1)$-fold fibre product and a predictable rank. A single failed rank prediction refutes the framework; each success is strong structural evidence.

9.4 What is proved, what is standard, what is a gap

Proved here or standard: Everything in Sections 2–7. Proposition 2.5, Proposition 3.3, Theorem 3.6, Proposition 9.1 are self-contained. Theorems 4.5, 6.2, 7.5 and Dubrovin–Guzzetti are literature.

The gaps:

1. **Order vs. argument.** Proposition 9.1 puts x in the Bessel *order*; the lattice lives at integer order and argument $\frac{1}{2}$. The missing step is a Kontorovich–Lebedev/stationary-phase argument showing that the x-integration in c_{mk} localizes the continuous-order kernel onto the two integer-order periods at the critical value. The appearance of $K_0(\frac{1}{2})$ in the closed form of c_{00} via Tricomi's $U(\frac{1}{2}, 1, \cdot)$ (`meixner-pollaczek2.tex`) is strong evidence that this localization is exact, not asymptotic.

2. **The third Γ-factor.** The kernel is $\Gamma(\frac{1}{4} + ix)^2 \Gamma(\frac{1}{4} - ix)$ — three factors, not a balanced pair. Three factors suggest a fibre product $(\mathbb{G}_m^2, u + v + \text{constraint})$ rather than the bare Bessel geometry; the reduction to rank two must come out of the computation rather than be assumed.

3. **The normalization $\sqrt{2e}$.** $e^{1/2} = e^{f(1)}\, e^{-f(-1)} \cdots$ — the factor is suggestively $e^{\text{critical value}}$, i.e. the regularization constant that relates thimble periods to Stokes-normalized flat sections, but we have not pinned it down.

We assess the probability that Conjecture 9.2 is correct as it stands at roughly 40%, and the probability that the correct explanation lives in this circle of ideas (exponential periods of a $\mathbb{G}_m$-geometry with a K-theoretic integral structure) as substantially higher. It is the only mechanism known to us that produces rank-two lattices with Bessel generators, quadratic relations with rational $2\pi i$-determinants, and Γ-factor kernels, simultaneously and for structural reasons.

Glossary of the main objects

object	role in the story
$\mathbb{G}_m$, $f = \frac{1}{4}(u + u^{-1})$	the stage; Mellin's half-line $\mathbb{R}_+ \subset \mathbb{G}_m$ is a thimble
$H^1_{\mathrm{dR}}(\mathbb{G}_m, f)$	rank 2 = #critical points; basis du/u, $u\,du/u$
$H_1^{\mathrm{rd}}(\mathbb{G}_m, f; \mathbb{Z})$	the integral lattice: thimble $\mathbb{R}_+$ and circle $\lvert u\rvert = 1$
period pairing	$\int_\gamma e^{-f}\omega$; values $2K_\nu(\frac{1}{2})$, $2\pi i(-1)^\nu I_\nu(\frac{1}{2})$
$\det P = -8\pi i$	Poincaré duality = Bessel Wronskian at $z = \frac{1}{2}$
$QH^*(\mathbb{P}^1) = \mathbb{C}[h,q]/(h^2 - q)$	one curve through two points; two vacua
quantum connection	$\hbar q\partial_q - h*$; QDE = modified Bessel equation
J-function $I_0(2\sqrt{q})$	curve counts; K is the second, transcendental solution
mirror map, $q = \frac{1}{16}$	critical values $\pm\frac{1}{2} = \pm 2\sqrt{q}$; the Bessel argument
$K^0(\mathbb{P}^1)$, $\widehat{\Gamma} = \Gamma(1+h)^2$	the $\mathbb{Z}$-structure; why Γ-kernels give integer matrices
Euler/Stokes matrix $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$	Dubrovin: derived category controls the irregular connection
$u \mapsto u^{-1}$	deck involution = functional equation $s \mapsto 1 - s$, $F \leftrightarrow G$
Weil rep. $\omega = \omega^+ \oplus \omega^-$	lowest weights $\frac{1}{2}, \frac{3}{2}$ = MP parameters 2λ; Mellin = $K \to A$ intertwiner

References (informal)

Bloch–Esnault, *Homology for irregular connections*; Givental, *Homological geometry and mirror symmetry / Equivariant Gromov–Witten invariants*; Iritani, *An integral structure in quantum cohomology and mirror symmetry for toric orbifolds*; Katzarkov–Kontsevich–Pantev, *Hodge theoretic aspects of mirror symmetry*; Dubrovin, *Geometry of 2D topological field theories*; Guzzetti, *Stokes matrices and monodromy of the quantum cohomology of projective spaces*; Fresán–Sabbah–Yu, *Quadratic relations between Bessel moments*; Kontsevich–Zagier, *Periods*.